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IoT based Distributed Framework for Agricultural Decision Support System

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Abstract—The integration of technology has revolutionized various industries of different domains, including agriculture. While agriculture is vital to many developing countries, reliance on traditional farming and the absence of data-driven methods have led to decreased productivity. Adopting new technologies is crucial for enhancing agricultural productivity and informed decision-making. When integrating these technologies, establishing standardized best practices in agriculture becomes essential. Thus, we propose the Internet of Things-based Distributed Agricultural Decision Support System (ID-ADSS), a data-driven, distributed framework. The framework uses an IoT-based decision-support system as a blueprint for developing agricultural systems or related infrastructure. ID-ADSS includes an entire ecosystem of data acquisition, processing, and analysis leveraging the concept of the Internet of Things (IoT), Low Power Wide Area Networks (LPWAN), and additional plugins such as machine learning technology, which are the best fit for agricultural data acquisition and inference systems. The framework focuses on distributed hierarchical layering to optimize the decision-making process by prioritizing regional parameters and considering the diverse agricultural geography. The ID-ADSS lays the foundation for the design and development of a comprehensive agricultural decision ecosystem. In addition, the conceptual framework is implemented and validated in a simulation, with LoRaWAN and other technologies.

Index Terms—Agricultural decision support system, Internet of Things, Low Power Wide Area Network, precision farming

I. INTRODUCTION

Agriculture serves as a significant source of income and sustenance for many regions. The limited adoption of modern technologies and persistence of traditional subsistence farming practices have led to diminished agricultural output, and a struggle to meet growing domestic consumption demands. To enhance crop efficiency and overall productivity, contemporary technologies such as Internet of Things (IoT), cloud-based systems, machine learning, and various wireless connectivity solutions can be adopted. In this context, the development of a suitable standardized framework for smart agriculture, encompassing data acquisition, processing, and information exchange, emerges as essential for the advancement of data-driven farming systems. The framework must exhibit the flex-ibility to accommodate a wide range of cutting-edge technolo-gies, including IoT, cloud computing, machine learning, Low Power Wide Area Networks (LPWAN), and more. The system, based on this framework, should automate the agricultural pro-cess, utilizing historical data to provide recommendations that enhance productivity. Farmers should possess the capability to analyze and monitor their farms, enabling them to make more precise decisions. This research presents an IoT-based Distributed Agricultural Decision Support System (ID-ADSS) incorporating the aforementioned technologies. This research illustrates the idea of this distributed hierarchical concept of a decision-support ecosystem for agriculture. The central idea of the framework is to effectively utilize a distributed approach, employing a well-structured hierarchy of layers to manage and organize information at varying levels of granularity, aligning with the specific requirements of diverse regions and localities. Since the parameters of agricultural decision support systems may vary between different geographical regions, the framework should be sufficiently capable of capturing the regional context to provide accurate decisions. Agricultural data often originates from distant farmlands, requiring data-collecting devices to send information over vast distances using minimal

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power. The framework has aligned these use case scenarios with relevant technologies for it to be efficient and practical. IoT is highlighted as a crucial technology in this framework as it allows for real-time monitoring and analysis of farm conditions. Additionally, the utilization of LPWAN technology within the IoT ecosystem enables low-power, long-range data transmission, making it well-suited for collecting data from remote farmlands, ensuring that even distant regions can benefit from data-driven farming solutions.

II. BACKGROUND AND RELATED WORKS

The concept of data-driven farming is relatively new to many countries as the adoption of such novel technologies in any domain is challenging due to various factors. These factors include governmental policy and priority, economic condition, the geography of farmlands, awareness among the policy-makers and farmers, brainstorming and public discussion on the topic, scientific research, publications, etc. There have only been a few research studies before regarding agricultural IoT in some specialized domains, however, research on the standardization of these systems using a conceptual framework is still lacking.

M. Shrestha and S. Khanal in their work [1], emphasized the critical importance of applying precise agricultural parameters that account for the spatial and temporal variability within fields. Their study revolves around the concept of precision agriculture and its significance in optimizing productivity. By utilizing their conceptual framework, they demonstrated how the precise management of agricultural inputs, taking into account the inherent variability in space and time, can lead to enhanced productivity in the field of precision agriculture. However, it is important to note that their work lacks compre-hensive implementation details of the proposed framework. The implementation of the framework is not elaborated. Fur-ther research is necessary to bridge this gap between theory and practical implementation in precision agriculture.

K. Haseeb, I. Ud Din, A. Almogren and N. Islam proposed a framework for smart irrigation in their paper, ”An Energy Efficient and Secure IoT-Based WSN Framework: An Applica-tion to Smart Agriculture”. They proposed an energy-efficient and secure network for the WSN framework. The framework employs techniques for a secure and energy-efficient system but lacks a general framework incorporating the various needs required for a complete agricultural ecosystem [2].

G. Gautam et al. proposed a framework for water supply management in Nepal in their research, “Usage of IoT frame-work in water supply management for smart city in Nepal”. The author proposed a framework for a water management system that aids in the development of smart cities in Nepal. The research proposes a framework for water management and also their implementation and experimental findings. The framework has been conceptualized and implemented in the water management system but it can’t be used for the diverse agricultural needs [3].

S. J. S. Prasad et al. [4] developed a private server that collects and transmits agricultural sensor data to the cloud without the use of IP-based routers. The authors claimed that the network interface is fulfilled by LoRa technology which solves the issue of communication failure problems and energy consumption. Although they designed and developed a network for agricultural IoT systems, full implementation, and customization are still missing.

A survey of different LPWAN Technologies such as LoRa, NB-IoT, Sigfox, and LTE-M in the agricultural field for India is presented by M. L. Liya, D. Arjun [5]. They claimed that LoRa is the best-suited technology among the compared technologies for the outdoor wide-area network of agricultural fields. The research illustrated the significance of various LPWAN technologies but a standard framework has yet to be conceptualized.

A. Azari and C. Cavdar modeled the LPWA-IoT network and performed various performance analyses like battery-lifetime analysis, and operation control of such networks. The author analyzed the system performance in different dynamic conditions by varying the density of the access points, communication bandwidth, volume of traffic, and QoS in com-munication. The implementation feasibility of an agricultural IoT infrastructure is justified by the performance of LPWAN networks with IoT [6].

M. T, K. Makkithaya et al. [7] proposed a federated learning method for training soybean yield prediction models on a horizontally distributed dataset located on different client devices. They used the FedAvg algorithm to train the ResNet-16 and ResNet-28 networks to predict soybean yield, and they compared the outcomes to those of conventional deep learning and machine learning techniques. In comparison to centralized learning models, they observed that federated averaging using the ResNet-16 regression model with Adam optimizer produced the best results. Such federated learning can be implemented in various nodes and servers to yield better results while distributing the computational needs. Also, federated models can be used for geographical region-based inference.

III. METHODOLOGY

The novelty of the proposed conceptual framework lies in its distributed hierarchical approach. The primary design phi-losophy involves utilizing this framework for data collection, processing, and inference to support decision-making. The

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hierarchical design caters to decision-support needs at varying levels of detail. The framework's design accounts for diverse agricultural landscapes and regional farming practices. Further, an agricultural system is simulated and analyzed based on the reference of the proposed framework.

A. Design of Distributed Framework

The framework includes a hierarchy of servers, consisting of a knowledgebase, inference engine, dashboard UI, and communication modules. Wireless sensor nodes (WSN) deployed in the farmlands are responsible for collecting various agricultural data in real time. Figure 1 shows the top-level conceptual design of the framework.

The base of the framework comprises various end sensor de-vices. These end devices are IoT devices capable of capturing and transmitting data using LPWAN technology. In addition to the acquisition, the end devices are also capable of edge com-puting and facilitating efficient automated tasks to optimize the farming process. A local server collects data from various end devices. These servers are responsible for collecting data from the end devices in a specific geographical area via gateways. This helps provide area-specific information and optimize the decision-making process within the given locality. Various local servers communicate and transmit data to a regional server. These servers operate in a large geographical area and contain the knowledge base for an entire region. The regional servers can be used by the regional or provincial government and stakeholders to help in their decision process. The central server aggregates data from all regions to provide a global per-spective. It provides a holistic view of the agricultural status of the entire country. This can be used by the central stakeholders to make a common decision using the nationwide scenario. To achieve the conceptual framework, a technological stack has been proposed in Figure 2. The end device is capable of data sensing and collecting agricultural parameters as well as edge computing. These devices communicate with local servers using Wireless LAN and message queuing services such as MQTT [8], Kafka [9], etc. Local servers aggregate and transfer data from end devices to regional servers, utilizing cellular networks, wireless gateways, or IP-based networks. The regional servers collect the data from various local servers. They transmit the data to a central server using IP-based internet connectivity. All these servers have communication modules, including networking and data queuing pipelines. The server also has its knowledge base, which is aggregated as we move up the hierarchy. Expert Systems such as rule-based, statistical inference models, etc. are also implemented on each of the servers. Different dashboards can be developed that act as a medium for question answering, communicating critical information, and providing the necessary alerts.

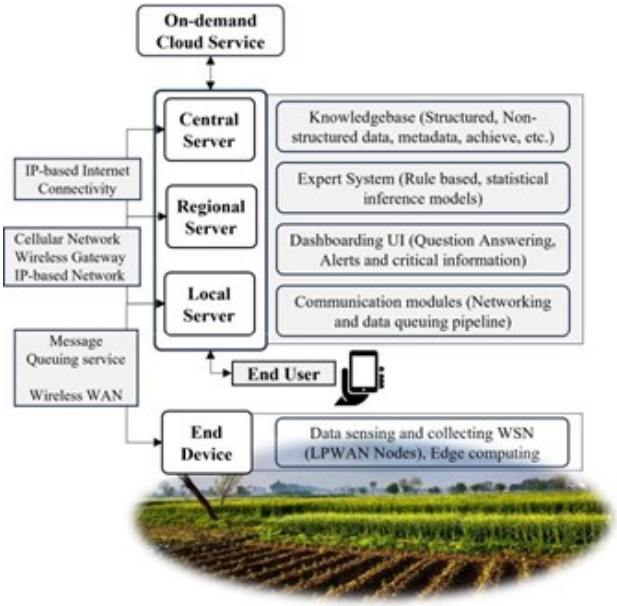

Fig. 1. Conceptual hierarchy of distributed ID-ADSS

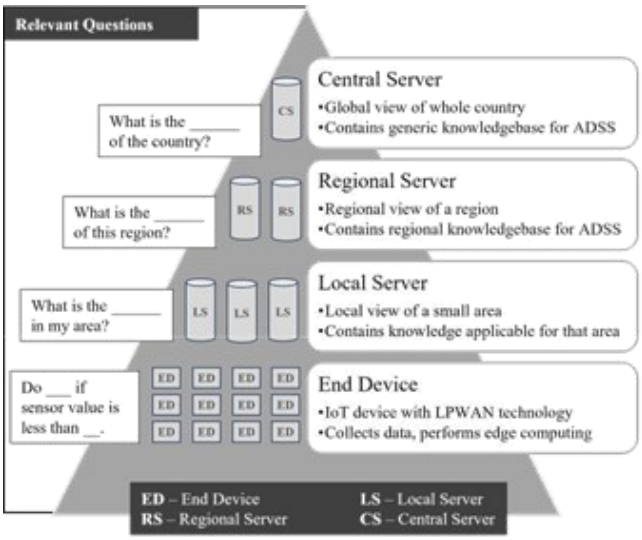

Fig. 2. Framework for the technological stack of distributed ID-ADSS

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The entire server is managed in the cloud and can be extended as needed. End users can also access the data on the servers based on their authorization and access roles. The users can be individual farmers, local governments, central governments, and other stakeholders, and role-based access is given to the specific set of servers and their data and system functionality.

1) Topological hierarchy of the ID-ADSS Framework: The topological hierarchy of the ID-ADSS Framework is shown in Figure 6. There can be many edge nodes situ-ated in agricultural farmlands. These nodes collect various data and properties of the specific farms which include soil properties such as pH, Nitrogen, Phosphate, Potassium, etc., meteorological parameters such as temperature, precipitation, and humidity, etc. and geographic parameters such as region, altitude, latitude, longitude, etc. These parameters are used by the edge nodes to perform various automated tasks to optimize the farming process. Various edge nodes are connected to one of the many local servers. These local servers can be common for a general area such as a ward or locality. The allocation of the local server is not rigid and can depend on the number of edge nodes under the local server. Similarly, the region server aggregates the data from several local servers. The region server can be assigned to Districts, or Province as per the number of local servers assigned to each. An IP-based network then aggregates the data from each of the regional servers onto a single central server.

Due to the low energy requirements and extended trans-mission range of LPWAN, the edge node in the topology employs this technology. The selection of a specific technology depends on various factors and requirements of the system. Table I summarizes a few major LPWAN technologies as a foundational guide for selecting a suitable technology.

2) Components of ID-ADSS Framework: The components of the ID-ADSS framework have been divided into three parts, i.e., core components, tools and services, and data sources as shown in Figure 3. Core components are the backbone of the ID-ADSS Framework. These components allow stakeholders to analyze the data and provide a medium to make efficient decisions. They encompass a knowledge base, expert systems, and machine learning models. Additionally, the framework integrates various tools and services. These tools allow the entire chain from local stakeholder to central, to optimize all their tasks. It includes crop modeling, pest or disease management systems, sustainability and environmental impact analysis, supply chain management, and so on. Data sources and their integration are fundamental components of the framework, which is sourced from both primary and secondary sources. The majority of the data is collected from sensors and laboratories while others can be collected from secondary sources such as news and media.

TABLE I.
LPWAN TECHNOLOGIES COMPARISON [10], [11]

	Unlicensed Spectrum		Licensed Spectrum	
	Sigfox	LoRaWAN	NB-IoT	LTE-M
Data rate (bps)	100	1760	50 k	1000 k
Duty cycle / Tx restriction	140 msg per day	1%	-	-
Battery life (Years)	7 to 8	8 to 10	7 to 8	10
Rural Range (Km)	30-50	15-50	20-40	50
Urban Range (Km)	3-10	2-15	1-5	10
Interference immunity	Low	Very High	Low	Low
Private Network Support	No	Yes	No	No
Latency (Second)	1	10	1.5-10	0.05-0.1

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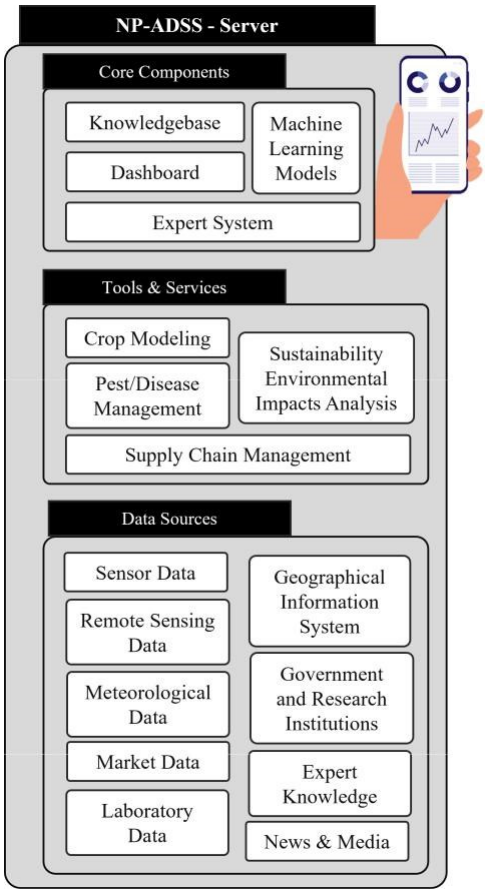

Fig. 3. Components of ID-ADSS Framework

Implementing the ID-ADSS framework involves deploying end devices with wireless sensors in farmlands, installing gate-ways, and designing software stacks or pipelines to process the obtained data on servers at various levels. Various existing tools and technologies can be used to implement these core components of the framework. Some of the technologies are listed in Table II. The table provides a foundational guide for choosing technologies. However, given the vast array of technologies in the market, the list is not all-inclusive. The requirements of a specific system will determine the most suitable technologies.

The utilization of a standardized reference architecture offers conceptual clarity and guides the selection of advanced technologies, resulting in the design of an efficient system that ultimately enhances agricultural performance. The hierarchical framework, with its focus on regional parameters, further enhances system performance by tailoring recommendations to the specific agricultural geography, ensuring optimal outcomes in diverse contexts. Furthermore, the ID-ADSS enhances decision-making with a standardized reference for efficient real-time data aggregation, a hierarchical architecture for re-gional decision support systems, and technology recommenda-tions. This promotes data-driven choices, improving precision in agriculture.

B. Implementation of Proposed Framework

To validate the feasibility of the proposed framework, an illustrative system has been designed and simulated using a simulation tool called OMNET++ [12]. For the simulation, a use case for Nepal has been taken. Considering the ge-ographical topography, three regions have been considered. The detailed block diagram of the implementation is shown in Figure 4. Since this section aims to demonstrate an exam-ple implementation of the proposed framework, no specific algorithm or functionality is implemented on the servers or nodes. Automatic irrigation and crop yield prediction are taken as examples of use cases in the block diagram. However, the implementation employs the logical layers suggested by the proposed conceptual framework to collect data from edge sensor nodes. Furthermore, the energy consumption pattern is analyzed to validate the energy feasibility of end nodes in farmlands. The implementation uses edge nodes to collect various parameters such as pH, NPK, and other soil-related parameters. The edge node then transfers the data to the nearest gateway, which acts as both the local as well as regional server in this case. The network server and central cloud server operate together to form the central server. The servers can be equipped with various algorithms and tools to analyze the data.

1) System Overview: The association of the framework and the example implementation is shown in Table III.

In the implementation, a total of 20 LoRa sensor nodes are placed in farmland all over Nepal as shown in Figure 5. The soil-related data collected by the nodes is forwarded to the nearest gateway server using either MQTT [8] or LoRa radio [13]. The gateway server modifies the LoRa-based packet

into a standard IP-based packet and transmits it to the nearest cellular base station. Then the packet is forwarded through the existing cellular or other IP-based infrastructure. Finally, the packet is routed to the corresponding application server via Network Server. The Network Server implements the LoRaWAN protocol, verifies the device's integrity and authen-ticity, and deduplicates uplinks [14]. It enables connectivity, management, and monitoring of devices, gateways, and end-user applications. For the small-scale simulation, local and regional servers are combined. There are three local/regional servers as wide area networks i.e., Western WAN, Mid WAN, and Eastern WAN. The corresponding local/regional gateway aggregates the data

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from each sensor node and combines them into a global. The implemented network topology is shown in Figure 5.

TABLE II
TECHNOLOGIES FOR CORE COMPONENTS IMPLEMENTATION

Component	Implementation Type	Technology	Use Case Examples
Knowledgebase	NoSQL	MongoDB	Capable of storing dynamic agricultural data
		Cassandra	Suitable for storing high-volume, time-series IoT data from farms
		Redis	Cache frequently accessed agricultural data for quick retrieval
	SQL	MySQL	RDBMS for structured data storage
		PostgreSQL	Advanced RDBMS for geospatial data storage and analysis
		SQL Server	Robust RDBMS for data warehousing and advanced analytics
Machine Learning	Supervised	Random Forest	Crop yield prediction based on multiple features
		Support Vector Machines	Classification like detecting crop diseases from sensor data
		Linear Regression	Predicting crop yields
		Convolutional Neural Networks	Disease classification based on image of plants
	Unsupervised	K-Means Clustering	Group plants based on similar sensor data patterns
		Principal Component Analysis	Dimensionality reduction and feature extraction from sensor data
		Isolation Forest	Detect anomalies or outliers in sensor data
	Reinforcement	Deep Q-Network	Scheduling irrigation based on weather change
		Proximal Policy Optimization	Optimizing control policies in automated irrigation management
		Deep Deterministic Policy Gradient	Tasks like precision agriculture equipment control
Expert System	Rule-Based Systems	Rule Engines	Diagnosis and treatment recommendation for crop diseases
	Knowledge Graphs	Resource Description Framework	Integrate and represent data from diverse sources semantically
		Web Ontology Language	Defining and reasoning for smart decision making
	Fuzzy logic based	Fuzzy Logic	Irrigation decisions in agriculture based on imprecise sensor data
	Case-Based Reasoning	CBR frameworks	Crop Disease Management based on historical outbreaks
Dashboard	Web based	HTML,CSS,Javascript(frameworks)	Interactive dashboard for real-time monitoring and visualization
		Node.JS, Django, Laravel	Backend for data manipulation and processing
	Mobile	React Native, Flutter, Swift, Kotlin	Mobile app control and monitor the farms
	IoT Dashboard	Python, C, C++, AWS IoT	Local control and monitoring of connected sensors & actuators
	Data Visualization	Tableau, Power BI, D3.js	Creating data-rich dashboards for data analysts and researchers
	Real-time	Apache Kafka, Flink, Spark	Monitoring real-time data and making immediate decisions
	GIS dashboard	ArcGIS, QGIS, Leaflet, Mapbox	Visualizing geospatial data on interactive maps

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2) Dataset collection: For the simulation, the data is taken from "Nepal Seed and Fertilizer Project (NSAF)" maintained by the CIMMYT and NARC. This tool provides different soil properties such as texture, structure, consistency, mottles, pH, color, slope, etc. Texture, pH, Total N, Available P, Available K, Boron, Zinc, and Organic matter were analyzed from physical and chemical analyses of soil samples at the soil lab [15]. The tool has the soil mapping data for the various regions in Nepal. For the simulation, data for each district is extracted by using a polygon select tool and selecting the area of the district manually. This process is repeated for all the districts to collect the entire dataset.

3) Energy model of nodes: To assess the feasibility of deploying LoRaWAN sensor nodes to remote farmlands where active forms of electricity may not be available, it is important to analyze the energy consumption of the LoRa node. A LoRa node operates in either sleep or active mode. During the sleep mode, power PSleep is consumed for unit time duration. In the active mode, the LoRa node can go through various states which include waking up, measuring the data, processing the data, waking the transceiver, transmitting data, and receiving data. The energy consumption pattern of the actual LoRa is modeled for the simulation purpose. Total consumed energy ETotal at end sensor node for one cycle is given by [16]:

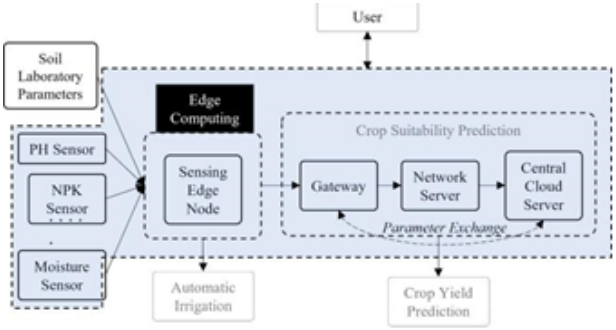

Fig. 4. Implemented System Block Diagram

TABLE III

FRAMEWORK IMPLEMENTATION

Framework	Example Implementation
LPWAN Technology	LoRaWAN
End Node Inputs	pH, NPK, Moisture, etc.
End Device	LoRa Sensor Nodes
Local Server	Server with LoRa Gateway
Regional Server	Server with LoRa Gateway
Central Server	Network Server, Central Cloud Server
Message Queuing	MQTT [8]
Wireless Technology	LoRa Radio [13]
No. of end nodes	20
No. of regions	3

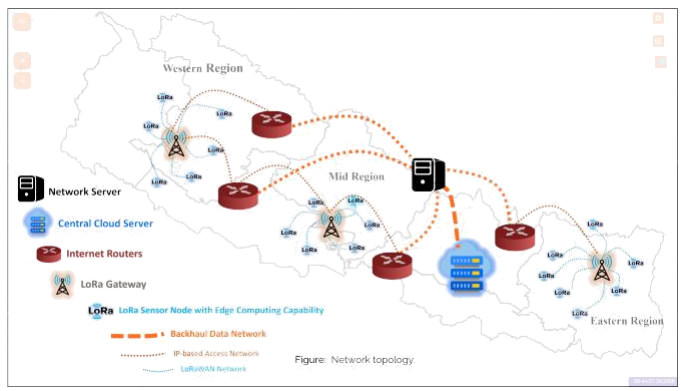

Fig. 5. Implemented Network Topology

$$E_{Total} = E_{Sleep} + E_{Active}, \quad (1)$$

where ESleep is energy dissipated by the sensor node in the sleep and EActive is energy dissipated during the active mode.

$$E_{Sleep} = P_{Sleep} \cdot T_{Sleep}, \quad (2)$$

where PSleep is power consumption and TSleep is the time duration in the sleep mode, respectively.
The total energy consumption EActive is given by:

$$E_{Active} = E_{WU} + E_m + E_{proc} + E_{WUT} + E_{Tr} + E_R, \quad (3)$$

where,
E_{WU} is the consumed energy during system wake-up, E_m is the consumed energy during data measurement,
E_{proc} is the consumed energy during processing,

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E_{WUT} is the consumed energy during the wake-up of the LoRa transceiver, E_{Tr} is the consumed energy during a transmission mode, E_R is the consumed energy during reception mode.

IV. RESULTS AND ANALYSIS

To demonstrate the implementation feasibility of the ID-ADSS framework, a LoRaWAN-based infrastructure is de-signed, implemented, and tested. The implementation follows the framework as listed in Table III. The various layers of the ID-ADSS framework are implemented in the simulation and their working is analyzed and validated. On the basis of the framework, the infrastructure in the simulation includes edge nodes that collect various agricultural parameters. These nodes send packets to a gateway, which then forwards them to the IP-based network. The central server subsequently collects these packets for further analysis. In the simulated environment, all the modeled components mimic the behavior of actual components.

A. LoRaWAN Network

The LoRaWAN topology has been implemented in the om-net++ network simulator. The LoRa sensor node is connected to the gateway via LoRa radio and the gateway connects to the central network server via an IP-based router device. Here, the sensor nodes represent the edge nodes/end devices that communicate with the local/regional server, i.e. gateway. The local/regional servers then connect to the central server. The LoRaWAN is simulated with the LoRa node parameters shown in Table IV. In the simulation, any end node can transmit its measurement data asynchronously. The transmission range of the node depends on various parameters such as transmitting power, spreading factor, and environmental interference. In the simulation, these parameters are controlled to adjust its transmitting range. Figure 7 shows the working example of a simulation where a node is transmitting a packet to all other nodes in its transmission range.

TABLE IV
LORA NODE PARAMETERS

Parameter	Value
Transmission Power	14 dBm
Carrier Frequency	868 MHz
Spreading Factor	12
Bandwidth	125 kHz
Coding Rate	4

Twenty LoRa sensor nodes and three regional gateways are chosen for the simulation as shown in Figure 7. For instance, LoRaNode[9] broadcasts a packet to all other nodes including gateways. Here a gateway closest to the broadcasting node will receive the packet. Since there are three regional gateways, the broadcasting of packets in the corresponding regions is shown in Figure 8, 9 and 10.

As mentioned in the framework, the edge nodes commu-nicate with the local/regional servers using MQTT or LoRa radio. The 3 regional servers aggregate the data from their edge nodes and then use an IP-based network to transfer the data to a central server as mentioned in the framework. Further, each of the servers can implement the various core components mentioned in the ID-ADSS framework.

1) Network Performance Analysis: Since the LoRa nodes are deployed in remote farmland where electricity may not be available, it is worth analyzing the energy consumption by the LoRa nodes. A node has been taken as a sample, and the energy consumption by the node is shown in Figure 11. The designed network is simulated for 10,000 seconds. The interval between successive packet generation was exponential distribution centered around 1000 seconds. A total of 13 packets were transmitted during the simulation interval. In the Class A mode of operation of the LoRa node, the LoRa protocol cycles through various states as defined in Section III-B3. These multiple states consume different energy levels.

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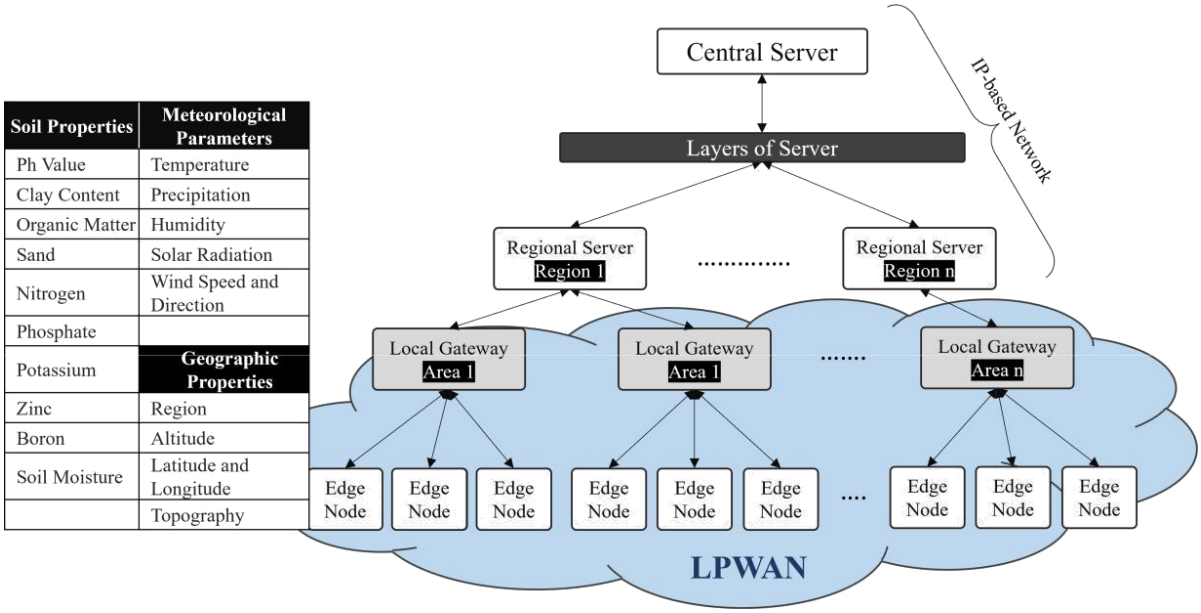

Fig. 6. Topological hierarchy of ID-ADSS Framework

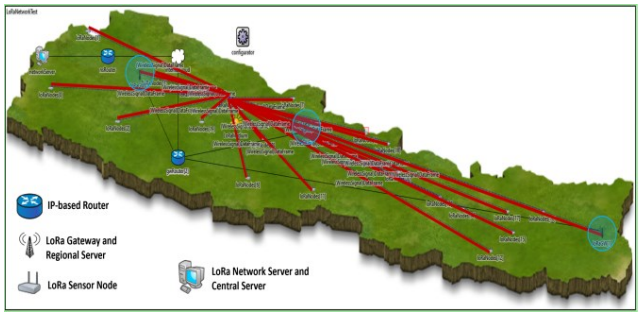

Fig. 7. Broadcasting a LoRa packet to all the nodes

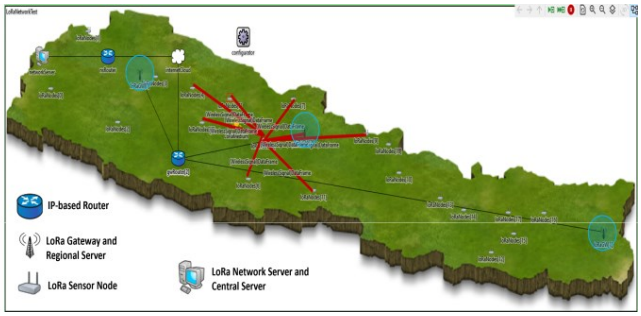

Fig. 9. Broadcasting a LoRa packet in Mid Region

Fig. 8. Broadcasting a LoRa packet in Western Region Region

Fig. 10. Broadcasting a LoRa packet in Eastern

In Figure 11, each vertical descent represents the energy drop during transmission. The node consumes very little energy (~ µJ) when it is in sleep mode. Assuming 0 as the initial value, the total energy consumed at the end of the simulation can be seen to be -3.852 J.

From the simulation, it is evident that the architecture is functioning. Each of the edge nodes is collecting data from the sensors and converting them to the necessary format. Further functionalities for edge computing and intelligent irrigation can be added. The data packets from each node are aggregated to the local/regional server. The servers then receive, combine, and process the data for further transmission.

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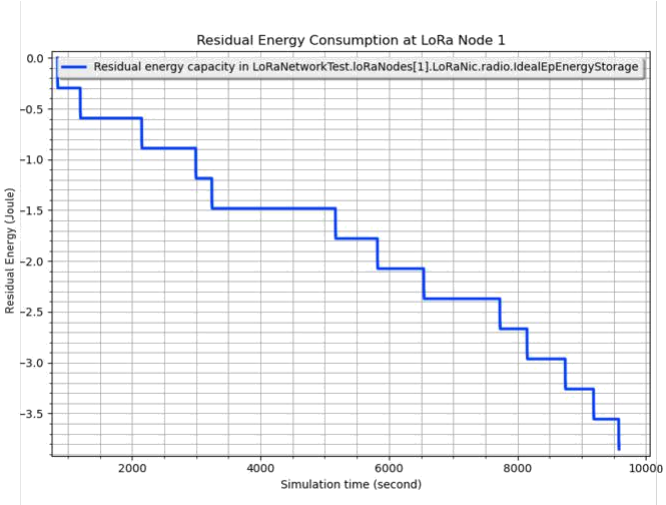

Fig. 11. Energy Consumption by a sample LoRa Node

TABLE V
SIMULATION SETUP AND RESULTS

Parameter	Value
LoRa node class	Class A
Total simulation duration	10,000 seconds
Packet generation interval	Exponential distribution centered around 1000 seconds
Total packets sent	13
Total energy consumption	3.852 Joule

to the central server. The aggregated data can be processed in the local/regional server by adding the components and tools mentioned. The central server collects all the data and processes it as required. Subsequently, the framework used for the design is feasible for the development of an entire ecosystem. Additional functionalities can be added to various layers as conceptualized in the ID-ADSS framework.

The proposed framework is compared with other similar architectures, as shown in Table VI. Given the conceptual nature of the framework, achieving a quantitative comparison presents significant challenges. Comparison is done based on the availability of various features such as standard framework, IoT implementation, scalability, etc. Upon comparison, the proposed framework excels above all and includes the features that are necessary for the development of any agricultural decision support system.

TABLE VI
COMPARISON WITH PREVIOUS WORKS

Parameter	Proposed	[1]	[2]	[4]	[6]
Standard Framework	Yes	Yes	No	No	No
IoT implementation	Yes	No	Yes	Yes	Yes
Energy Consideration	Yes	No	Yes	Yes	Yes
Scalable	Yes	No	Yes	Yes	No
Multilayered Approach	Yes	No	No	No	No
Extendable Technology	Yes	Yes	No	No	No

V. CONCLUSION

In conclusion, an IoT-based distributed framework for Agri-cultural Decision Support System (ID-ADSS) is designed. The framework follows a hierarchical approach consisting of a stack of intelligent computational layers. The hierarchical distributed approach provides more control over scalability and granularity while designing the actual system. The various tools and technologies associated with the implementation of the framework have also been discussed along with the core components. Furthermore, a system has been implemented following the proposed framework and the feasibility of the distributed ID-ADSS framework stack has been validated using the simulated system.

For future research, hardware experimentation in the real world can be conducted based on the framework to further validate its significance. This includes exploring areas such as optimizing sensor networks, specialized machine learning algorithms, energy-efficient IoT deployments, and integrating blockchain into the agricultural supply chain, among other potential areas of investigation.

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